Practice as Research

LECTURE 1 OF 8 — COURSE INTRODUCTION & "WHAT IS RESEARCH?"

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Why this course exists

You already know how to make things. This course is about making things that **mean something to a wider conversation.**

THREE CLAIMS WE KEEP RETURNING TO

1. **Finding a voice** — a unique approach others can read in multiple ways.
2. **Finding a problem** — locating a real gap in current understanding.
3. **The almost-perfect situation** — a gap that *also* lets you develop a voice.

Two routes to the same place

If you come from artistic practice

The work is about finding a voice — an approach, a sensibility, a way of asking questions through making.

If you come from a technical background

The work is about combining technologies, developing a new use of a technology, or co-opting one in an unexpected way.

The strongest projects do both.

What you will be assessed on (in broad terms)

I'm team-teaching, so others will set specifics. But across the field, the grading logic tends to be:

- **C / B band:** a project that draws inspiration from specific works, with clear attribution and competent execution.
- **A band:** a project that *builds on* existing work to produce a notably new or unique outcome.

Citation is the floor. Contribution is the ceiling.

The 8-week arc

WEEK	FOCUS
1	What is research? Why does it matter? Who is it for?
2	How practice produces knowledge (Smith & Dean)
3	Situating research in a field — finding the GAP
4	Choosing and designing methods
5	Reflection as research insight — annotated documentation
6	What to document and why
7	Exhibition vs research output — complementary, not opposed
8	Critical framing — refining your argument

Three habits, starting today

1. **Keep a working notebook.** Anything — text, sketches, links, code, voice notes. Date everything.
2. **Read one thing a week you wouldn't normally read.** Outside your discipline.
3. **Make something small every week.** A test, a sketch, a probe. Not a finished piece.

The students who do these three things produce the best work. Every year. Without exception.

Today's session

THREE QUESTIONS

1. **What is research?** (the edge, the hypothesis, the test)
2. **What is the point of research?** (new knowledge — and is knowledge fixed?)
3. **Who cares?** (who is research for, who benefits)

Then: **examples** — what does this look like in technology arts?

What is research? (a working definition)

Research is a structured attempt to produce something that wasn't known before, in a form others can examine, test, or build on.

THREE COMPONENTS

- **The edge** — where the known runs out.
- **The hypothesis** — your bet about what's beyond it.
- **The test** — what you'll do to find out, and how others will judge it.

"The edge"

The edge is **what is not yet understood**, or **what is understood but unexplored**, or **what is assumed but not tested.**

LOCATING THE EDGE REQUIRES

- knowing what is already done *(Lecture 3)*
- knowing who is doing it now *(Lecture 3)*
- knowing what *they* think is unresolved *(Lecture 3)*

You cannot find an edge from inside your own head.

Project vs Theme vs Research Question

	WHAT IT IS	EXAMPLE
Project	A thing you intend to make	"I'll build a generative audio installation"
Theme	A territory you're interested in	"Memory, sound, and place"
Research question	A specific, answerable interrogation of the territory	"Can a generative audio system trained on a specific location's recordings produce material that listeners associate with that place?"

Most students arrive with a project. The work of this module is to get you to a research question.

Hypothesis, in practice-based work

A hypothesis doesn't have to be a falsifiable scientific claim. In practice-based research it is more often:

- **A wager** — "I think if I do *this*, *that* will happen, and it will be interesting because…"
- **A position** — "I claim that current approaches miss *this*, and my work demonstrates an alternative."
- **A probe** — "I don't yet know what happens if *this*, and finding out is itself the contribution."

All three are legitimate. They map onto **experiment, prototype, and probe** — which we'll cover next week.

The test

What counts as a "test" in practice-based research?

- **Public exhibition** — does the work read as you intended? What do audiences do with it?
- **Peer review** — does the field accept the framing, the method, the contribution?
- **Critical writing** — can you articulate what you did, why, and what it shows?
- **Replication / extension** — can someone else build on it?

A piece of work that no-one but you can interrogate is not yet research.

It might still be art. It might still be valuable. But it is not research.

What is the point of research?

To produce new knowledge.

But — is knowledge a fixed thing?

- Knowledge as **propositional** ("X is true")
- Knowledge as **procedural** ("here is how to do X")
- Knowledge as **tacit** ("I know how to do X but cannot fully say how")
- Knowledge as **situated** ("X is true *here, for these people, in this context*")

Practice-based research mostly produces the **procedural, tacit, and situated** kinds. This is not a deficiency — it's the point.

Why does it matter?

Research in technology arts has, in the last 25 years:

- Shaped how we think about machine learning's creative potential — *before* the current AI moment.
- Built tools that millions of artists now use (Wekinator, Magenta, RunwayML, Stable Diffusion's UI ecosystems).
- Forced legal, ethical, and aesthetic questions onto policy agendas (copyright, training data, authorship).
- Produced cultural artefacts that change how people understand the technology itself.

If artists weren't doing this work, someone else would be defining what these technologies are for.

Who cares? Who is it for?

Research has multiple audiences. Be honest about which one(s) you're addressing:

- **Other practitioners** — they want methods, tools, and demonstrations they can build on.
- **Academic peers** — they want a clear contribution to a literature.
- **Audiences / publics** — they want experiences that move, provoke, or illuminate.
- **Industry / policy** — they want frameworks, evidence, and arguments.
- **Future you** — they want documentation good enough to build on in 2 years.

Most strong projects address 2–3 of these *deliberately*. Weak projects address none of them and hope for the best.

Some broad examples of art-tech research

A small, partial set of examples — there are significant gaps in this list. None of this is required reading today, but it gestures at the soil this course grows from.

- **Roy Ascott** — telematic art (1980s). Networks as artistic medium *before* the web. *La Plissure du Texte* (1983) explored distributed authorship across continents.
- **Nam June Paik** — video synthesisers, *TV Buddha*, *Magnet TV* (1960s–70s). Co-opting consumer technology to expose its assumptions.
- **Janet Cardiff & George Bures Miller** — audio walks, immersive sound installations. Research into perception, place, and narrative through built experience.
- **Rafael Lozano-Hemmer** — *Pulse Room*, *Vectorial Elevation*. Public-scale interactive systems. Research into participation and crowd phenomena.

Why those four?

Each of them did one specific thing this course is asking *you* to do:

- **Ascott** — situated his practice in a *theoretical* conversation (cybernetics, systems theory) and produced work that advanced both the theory and the art.
- **Paik** — took a mass-market technology and used it against its own grain. The work *is* the argument.
- **Cardiff & Miller** — built rigorous methods around an experiential medium that resists rigour.
- **Lozano-Hemmer** — made the engineering visible as part of the meaning, not hidden behind the spectacle.

Closer to home: Christian Marclay's *The Clock*

24-hour single-channel video, 2010. A working clock made entirely of film clips showing the corresponding time. Winner of the Golden Lion at the 2011 Venice Biennale.

- **The edge:** can found footage be reorganised so that the artwork *is* a functional object?
- **The method:** an enormous, carefully indexed database; bespoke software for synchronisation and playback; rigorous editorial decisions.
- **The contribution:** a new compositional logic for moving image — and a working argument about cinema, time, and labour.

I designed and built the playback software for the first version of *The Clock*. The technical work is part of the artwork, not a service to it.

Closer to home: Sigur Rós, *Route One* (2016)

A 24-hour generative live broadcast: a slow drive around Iceland's Route 1, with the band's music continuously regenerated by software in real time so that no two minutes are the same.

- **The edge:** can a piece of recorded music be re-imagined as a live, infinite, generative system — broadcast to a public audience?
- **The method:** custom generative audio software (which I wrote), long-form camera work, public broadcast infrastructure.
- **The contribution:** a working demonstration that "the album" and "the live performance" aren't the only formats music can take.

From my own practice: MAGNet (2016)

Neural audio synthesis at CD quality, using LSTMs over STFT frames. Among the first deep learning systems to generate raw audio with musical structure.

- **The edge (then):** generative audio was either symbolic (MIDI) or low-fidelity. Could deep learning produce *listenable* musical material?
- **The method:** training neural networks on spectral representations; iterative listening tests; collaboration with composers (Robert Del Naja).
- **The contribution:** a working architecture, public demonstrations (Barbican *AI: More Than Human*, 2019), and a body of subsequent work — including the *Network Bending* lineage which gave artists a vocabulary for *intervening inside* generative models.

Related: Memo Akten's *Learning to See* (2017, with Fiebrink & Grierson) and Terence Broad's *Blade Runner—Autoencoded* (2016) sit in the same lineage.

From a recent CCI student: Adam Cole, *Kiss/Crash*

Adam Cole's MSc work — a deeply researched immersive triptych engaging with how generative image models depict intimacy and violence, and what that says about their training distributions. Shortlisted for the Lumen Prize; XR Audience Award winner at SXSW 2024.

- **The edge:** what do these models *avoid*, *conflate*, or *hallucinate* when asked about contact between bodies?
- **The method:** systematic prompting, comparative analysis, careful framing of presentation — written up as a SIGGRAPH 2023 Art Paper.
- **The contribution:** simultaneously an artwork and a critical investigation of the model's latent geometry.

This is what an A-grade outcome looks like: voice + gap, executed.

A research lineage from one paper

Broad, Berns, Colton, Grierson (2021), *Active Divergence with Generative Deep Learning* (ICCC'21).

The paper surveys techniques that **deliberately push generative models away from their training distribution.** It cites:

- *Teratome* (Broad, 2020) — network bending as artistic method.
- *You Are Here* (Schultz, 2020) — chained models, custom GANs, super-resolution.
- Network blending (Pinkney & Adler, 2020) — combining models trained on different data.
- *Network Bending* (Broad, Leymarie, Grierson, 2021) — the underlying technique paper.

One good survey paper is worth a hundred Twitter threads.

What approaches do people agree are worthwhile?

There is genuine consensus in the field on a few things:

- **Reproducibility-where-possible.** Code, data, documentation that lets others extend your work.
- **Honest negative results.** "I tried X, it didn't work, here's what I learned" is publishable.
- **Open methods.** The how matters as much as the what.
- **Critical reflexivity.** What assumptions did you bring? What did the work surface that you didn't expect?
- **Ethical seriousness.** Especially around training data, labour, consent, and representation.

These are also the criteria that get work into the venues that matter — NeurIPS Creativity workshops, ISEA, Ars Electronica, Leonardo.

Where there is *not* consensus

Be aware these are contested:

- What "originality" means in an era of large pre-trained models.
- Whether AI-generated work can be authored, and by whom.
- Whether scale (compute, data) is itself a form of contribution.
- The relative weight of artefact vs documentation vs argument.
- Who counts as "the field" — academia, industry, festival circuits, online communities.

You don't need to resolve these. You do need to know they exist, and to position your work knowingly within them.

A simple working process (for the rest of the term)

Every week, for your own project:

1. **One thing read** — an article, paper, or work, with a paragraph of notes on what it does and what it leaves open.
2. **One thing made** — a sketch, prototype, test, or fragment.
3. **One question refined** — the research question you came in with, rewritten in light of (1) and (2).

Bring all three to your tutorials. By week 8 you will have 24 reads, 24 makes, and 24 versions of your question.

You will not be short of material.

Common failure modes

In my experience, the projects that struggle do one of these:

- **All theme, no question.** ("I'm interested in memory and sound.") — Becomes a mood board.
- **All project, no context.** ("I'm building a synth.") — Becomes a tools demo.
- **All technique, no purpose.** ("I'm using diffusion models.") — Becomes a tutorial.
- **All argument, no artefact.** ("I want to critique surveillance.") — Becomes an essay.
- **All artefact, no argument.** ("Here's the thing I made.") — Becomes a portfolio piece.

The good news: each of these is recoverable. We have 8 weeks.

What I want you to leave with today

01 Research is a structured attempt to make something not previously known, in a form others can examine.

02 Practice-based research produces procedural, tacit, and situated knowledge — which is real knowledge.

03 The difference between a project and a research project is whether you can name the edge, the wager, and the test.

04 Your assessment will reward citation as the floor and contribution as the ceiling.

05 The weekly habits — read, make, refine — are the actual work of this module.

For next week

Read **one** of these (we'll discuss in week 2):

- Smith, H. & Dean, R. T. (eds.) (2009). *Practice-led Research, Research-led Practice in the Creative Arts.* Edinburgh University Press. — Introduction chapter.
- Candy, L. (2006). *Practice Based Research: A Guide.* CCS Report, University of Technology Sydney.
- Borgdorff, H. (2011). *The production of knowledge in artistic research.* In Biggs & Karlsson (eds.), *The Routledge Companion to Research in the Arts.*

Bring **one** half-page note: what does this text say practice-based research *is*, and where do you agree or disagree?

Tutorials and contact

To be filled in based on team-teaching arrangements:

- Tutorial slot times
- Studio access / equipment
- Reading list location
- Discord / Slack / forum
- Office hours

Questions

What's unclear?

What worries you?

What are you already excited about?